

Emerging technological trends shaping up embedded development

1. Open Source platforms
2. Green technologies
3. Interoperability
4. Hardware standardisation
5. Wireless connectivity
6. Advanced user experience

and command higher average selling prices. But these exist in smaller unit quantities, making MCUs the largest dollar-volume segment of the four embedded processing technologies," adds Shailesh Thakurdesai, business development manager, microcontrollers, Texas Instruments India.

In MCU designs, there have been tremendous improvements in the past decade. MCUs are becoming more and more application-specific and product design engineers now have a lot of options to choose from. MCUs are becoming smaller with more RAM, Flash and peripherals on chip. Also, 32-bit MCUs based on ARM core are becoming popular and cost-effective because of the well connected ecosystem for semiconductor companies, software and hardware tool providers, design service providers and product design houses.

Another key trend in MCU design is the inclusion of digital signal processing (DSP) and floating-point processing unit in the controllers. Controllers with support for DSP are termed as digital signal controllers. These controllers deliver the performance required for DSP applications and the flexibility and ease of an MCU.

Moore's law has significantly pushed the processing capabilities of today's processors with multiple cores getting packaged into small silicon. Reconfigurable FPGAs are growing in size and providing capabilities such as SoC by incorporating processing cores, DSPs and inputs/outputs. Web- and cloud-enabled technologies are becoming very accessible, driving the systems to be more networked and intelligent. Hardware is becoming more of a

commodity and software is becoming the key differentiator of embedded products.

Some of the world's best and biggest silicon companies are fabless semiconductor companies. India needs to adopt a mature approach and focus on the maximum value creation, to build end-to-end products indigenously, without worrying about component-level indigenisation. This is similar to what has happened in the automotive industry – we started assembling automobiles using CKD kits and gradually moved to manufacturing components.

Areas of concerns for embedded engineers

Virtual platform or virtual prototype is the place where electronic design automation and embedded software converge. One of the major challenges faced by embedded engineers is with regard to the time taken for hardware simulation and modeling. There are also issues dealing with hardware dependencies and repetitive development of drivers/middleware for embedded systems.

"Today, over 70 per cent of integrated circuit design is on SoCs, which are a combination of embedded processors and software. However, for embedded systems engineers, system-level verification capabilities haven't advanced with increasing complexity in hardware and software components. To combine today's complex hardware with more sophisticated software, electronics companies need better verification capabilities," says Alok Jain, engineer, Cadence Design Systems (I).

A unified approach to system verification through automated test generation, system-wide management and high-performance engines is needed to tackle cross-domain verification and debug needed to achieve more complex system-level products in lesser time.

Roadmap for the industry

"While India is well positioned to capitalise on the addressable market

Embedded Systems Original Design and Manufacturing



Integrated Embedded Product Development

- Software, Hardware, Mechanicals
- Concept to Prototype to Production

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